

Micro MPCs and Macro Counterfactuals: The Case of the 2008 Rebates

Orchard, Ramey & Wieland (NBER WP 31584; QJE 2025)

Kacper · Macro Reading Group

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The question

- Leading household studies (Parker et al. 2013): 3-month MPC of **0.5–0.9** out of the 2008 rebate
- These estimates calibrate the whole HANK literature and justify large transfer multipliers
- Put them in a standard NK model: what aggregate path do they imply?

Claim of the paper

High micro MPCs imply *implausible* macro counterfactuals for 2008. Once you fix the estimates *and* the model, the GE consumption multiplier is below 0.2.

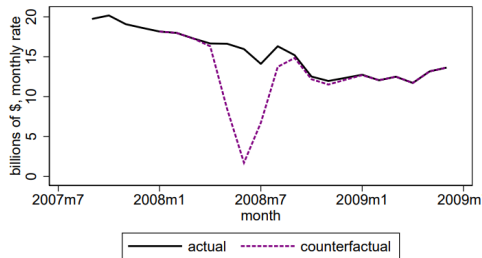
The method: macro counterfactuals as a plausibility check

1. Calibrate a standard model to the micro MPC
2. Simulate the rebate; subtract from actual data $\Rightarrow$ counterfactual “no-rebate” path
3. Ask: is that counterfactual believable, given forecasts and history?

A “no” means the micro estimate, the model, or both are wrong. Here: both.

The warning sign

Figure 1. Expenditures on New Motor Vehicles: Actual vs. Counterfactual



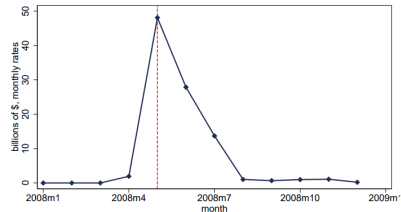
Note. Based on Sahm, Shapiro, and Slemrod calculations applied to revised data.

Absent the rebate, vehicle spending would have fallen from \$17bn (Mar) to under \$3bn (Jun), then rebounded — yet the Great-Recession low was \$12bn. Strains credulity. This paper puts that intuition into GE.

The 2008 rebate

- \$100bn ($\approx 11\%$ of monthly disposable income)
- 85% of households; average check \$1,000
- Distributed Apr–Jul 2008; timing random (last 2 SSN digits)
- Almost half paid in May alone

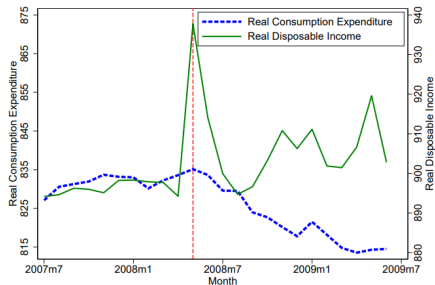
Figure 2. 2008 Tax Rebates



Notes. Data from Shapiro and Slemrod (2009). Rebates are nominal. Vertical red dashed line indicates May 2008.

In the aggregate: income jumps, consumption barely moves

Figure 3. Real Aggregate Disposable Income and Consumption Expenditure



Source: BEA data. Vertical red dashed line indicates May 2008.

Big spike in income, only a small bump in consumption. Feldstein (2008) and Taylor (2009) concluded the rebate did little - then were overshadowed by the micro studies. Not a smoothing artifact (retail sales, car sales, CEX all show no spike).

Two ways to reconcile micro with aggregate

1. The micro MPC estimates are **biased upward**
2. General equilibrium **dampens** rather than amplifies the micro MPC

The paper argues *both* are needed. Authors build the counterfactual, then fix both effects.

Baseline model: two-good, two-agent NK (TANK)

Ramey (2021) $\times$ Galí et al. (2007), monthly; durable good = motor vehicles.

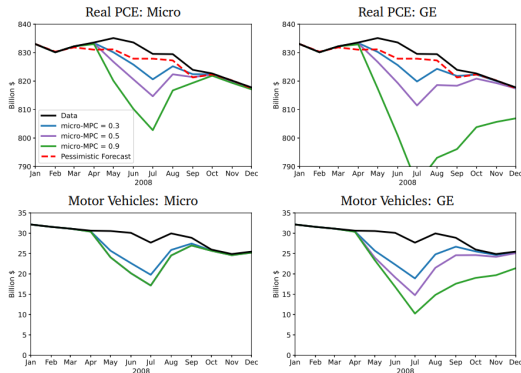
- Fraction γ hand-to-mouth; calibrate γ to hit the micro MPC (0.3, 0.5, 0.9)
- Durables and nondurables perfectly substitutable in production $\Rightarrow$ relative price fixed at $p_t^d = 1$
- Rich GE forces (Keynesian multiplier, sticky prices/wages, capital)

GE-MPC

Aggregate consumption response per \$1 of rebate = direct micro effect + induced Keynesian-multiplier effect.

Baseline counterfactuals: deep V-shapes, and GE amplifies

Figure 4. Counterfactual Real Consumption Expenditures: Baseline Model



Notes. Based on Two-Good, Two-Agent NK model simulations and actual data on rebates and consumption. The micro MPC value refers to the MPC for total consumption.

Flat Phillips curve + interest-rate inertia $\Rightarrow$ little real-rate crowding-out, so the Keynesian multiplier dominates and the counterfactual V is *deeper* in GE. Vehicle drop of 50–70% under Parker MPCs.

Is the V-shape believable? No.

- No 2008 forecaster (Greenbook, SPF, Goldman) predicted a summer collapse
- Their own *pessimistic* forecast (oil, recession, Lehman) still lies above all counterfactuals
- Actual – pessimistic gap = \$20bn $\Rightarrow$ $GE-MPC \leq 0.2$

Table 2. Model Counterfactuals Compared to Largest Historical Expenditure Decline

Panel A: Total PCE				
Largest Historical Declines			Model Counterfactuals	
Date	Episode	Decline	Calibration	Decline
Jan-Apr 2020	COVID lockdowns	17.4	micro MPC = 0.9	6.0
Jan-Apr 1980	Credit controls, Volcker	2.9	micro MPC = 0.5	2.7
Aug-Nov 1974	prior spike up	2.3	micro MPC = 0.3	1.6
Apr-Jul 1960	prior spike up	1.8		
Sep-Nov 2008	Lehman Collapse	1.1		
Panel B: Motor Vehicle Expenditures				
Largest Historical Declines			Model Counterfactuals	
Date	Episode	Decline	Calibration	Decline
Jan-Apr 2020	COVID lockdowns	31.2	micro MPC = 0.9	66.7
Aug-Nov 1974	prior spike up	25.3	micro MPC = 0.5	51.9
Jul-Oct 2005	prior spike up	25.3	micro MPC = 0.3	38.2
Jan-Apr 1980	Credit controls, Volcker	24.8		
Sep-Nov 2008	Lehman Collapse	16.9		

So either the micro MPCs are too high, or the model lacks GE dampening. According to the paper - both.

Reconciliation (1): three upward biases in the micro MPC

Standard TWFE / event-study regression (test of the PIH):

$$C_{i,t} - C_{i,t-1} = \sum_s \beta_{0s} \text{month}_s + \beta'_1 X_{i,t} + \beta_2 I(\text{ESP}_{i,t}) + u_{i,t}$$

Replication: full-sample MPC = 0.5, rebate-only > 0.8 — matches Parker et al.

Three separate biases, all pushing β_2 **up**.

Bias 1: omitted lagged rebate (Kaplan–Violante)

First-difference the true one-time-effect model $\Rightarrow$ the lagged rebate lands in the error term:

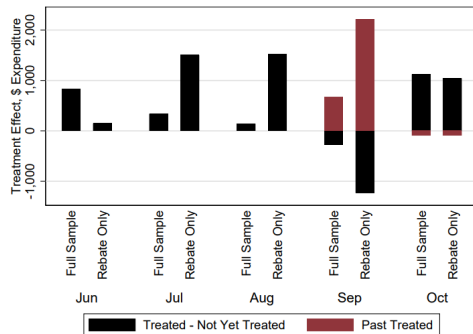
$$\beta^{OLS} = \beta - \beta \frac{\text{Cov}(\hat{D}_{i,t-1}, \hat{D}_{i,t})}{\text{Var}(\hat{D}_{i,t})}$$

- Staggered payout $\Rightarrow \text{Cov}(\hat{D}_{t-1}, \hat{D}_t) < 0$, so $\beta^{OLS} > \beta$
- Intuition: the t -treated are compared to the $t-1$ -treated, whose spending is *mean-reverting down*
- Valid PIH test, but the coefficient is *not* an MPC

Bias 2: forbidden comparisons (heterogeneous effects)

- A single lag removes only *average* mean-reversion
- OLS uses later cohorts (small reversal) to correct earlier cohorts (large reversal) — undercorrects
- If early effects $>$ late effects $\Rightarrow$ upward bias (Sun–Abraham, Borusyak–Jaravel)

Figure 6. TWFE Coefficients in the Full and Rebate Only Samples By Month



Bias 3: rebate reporting bias

- Reported receipt correlates with *low* prior spending: a large negative “future-rebate → current spending” coefficient (−\$863 full, −\$575 rebate-only)
- Households over-report receiving the rebate in the month before the interview (recall error) — reported timing is *not* random even though true (SSN) timing is
- With mean reversion, the post-report bounce is misread as the rebate ⇒ upward bias. Fix: control for lagged expenditure

Correcting all three: MPC falls to ≈ 0.3

Implied 3-month MPC	(1) base	(2) +lag	(3) +het	(4) +lagged exp
Full sample	0.50	0.46	0.37	0.28
Rebate-only	0.82	0.56	0.64	0.34

Table 3. Household PCE Response to Rebate

Panel A: Full Sample					Panel B: Rebate Recipients Only				
	Homogeneous Treatment		Heterogeneous Treatment			Homogeneous Treatment		Heterogeneous Treatment	
	(1)	(2)	(3)	(4)		(1)	(2)	(3)	(4)
Rebate Indicator	470.13** (213.56)	433.84** (206.72)	347.40* (211.03)	261.99 (182.84)	Rebate Indicator	764.46** (314.81)	527.34 (342.40)	601.27 (403.52)	320.72 (496.73)
Lag Rebate Indicator		-173.61 (222.27)	-82.52 (201.72)	-60.92 (167.69)	Lag Rebate Indicator		-428.09 (354.06)	-178.65 (319.11)	-351.86 (360.43)
Lag Total Expenditure				-0.26*** (0.03)	Lag Total Expenditure				-0.29*** (0.02)
Lag Motor Vehicle				-0.74*** (0.03)	Lag Motor Vehicle				-0.71*** (0.03)
Implied 3-month MPC	0.50	0.46	0.37	0.28	Implied 3-month MPC	0.82	0.56	0.64	0.34
Implied 6-month MPC		0.72	0.64	0.20	Implied 6-month MPC		0.65	1.09	0.02
6-Month MPC S.E.		(0.48)	(0.49)	(0.37)	6-Month MPC S.E.		(0.92)	(1.07)	(1.18)
Income Decile FE	No	No	No	Yes	Income Decile FE	No	No	No	Yes
Observations	16,962	16,962	16,962	16,962	Observations	10,076	10,076	10,076	10,076

Both round to **0.3**. Borusyak et al. (2022) imputation gives the same (≈ 0.2 full, 0.37 rebate-only).

And it's almost all motor vehicles

MPC (spec 4)	Motor vehicles	Other PCE
Full sample	0.30	-0.02
Rebate-only	0.26	0.08

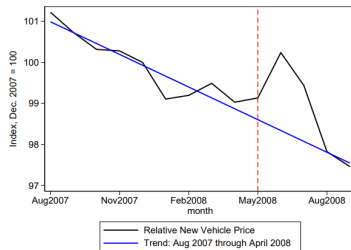
No support for the standard “0.25 on nondurables” HANK calibration. The entire response is durables.

But 0.3 in the baseline model is still implausible

With perfectly elastic durable supply, even a 0.3 micro MPC amplifies in GE. What's missing?

The relative price of vehicles.

Figure 7. Motor Vehicle Relative Prices



Source: BLS research CPI for new motor vehicles divided by core CPI (Williams et al., 2019).

≈1.5% above trend right after the rebate.

Reconciliation (2): upward-sloping durable supply

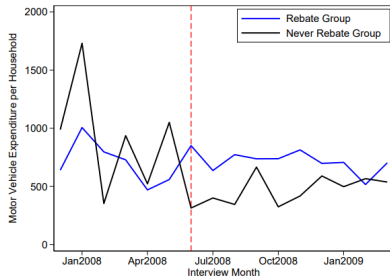
Following McKay–Wieland, let durables be produced with a fixed factor $\Rightarrow$ upward-sloping supply:

$$p_t^d = \left(\frac{X_t}{\bar{X}} \right)^{\frac{\zeta}{1+\zeta}}, \quad \text{supply elasticity} = \frac{1}{\zeta} = 5$$

- Hand-to-mouth pile into vehicles $\Rightarrow$ relative vehicle price rises (+1.1%, Jul 2008)
- Vehicle demand is elastic $\Rightarrow$ optimizing households are **crowded out**
- Calibrated to match the observed price spike

Direct evidence of crowding out

Figure 9. Motor Vehicle Spending per Household by Rebate Status



The control group is hit by the higher price too — consistent with the supply-side channel, not just a demand shock.

The payoff: micro MPC $\rightarrow$ GE-MPC

micro MPC	GE-MPC (total)	GE-MPC (vehicles)
0.30	0.07	0.09
0.50	0.22	0.15
0.90	1.42	0.45

At the preferred 0.3, dampening dominates. **7 cents per dollar**. Counterfactual now plausible.

Conclusion

- Big micro MPCs $\Rightarrow$ implausible V-shaped 2008 counterfactuals
- Fix the estimates: three biases $\Rightarrow$ micro MPC ≈ 0.3 , all vehicles
- Fix the model: elastic durable supply $\Rightarrow$ GE dampening
- Together: **GE consumption multiplier below 0.2**

A micro MPC is not an aggregate multiplier — and its *goods composition* decides whether GE amplifies or dampens.